EDS 223: Mapping Environmental Inequities

Potential Environmental Justice Concerns in Monterey County

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2026-10-20

Background

The Environmental Protection Agency (EPA) developed EJSCREEN, an environmental justice (EJ) mapping and screening tool that combines environmental and demographic indicators[1]. This analysis uses the EJSCREEN dataset to examine potential environmental justice issues in Monterey County.

These maps compare proximity to Risk Management Program (RMP) facilities with the distribution of people of color (POC) and limited English-speaking households. RMP facilities handle large quantities of extremely hazardous regulated substances and are required by the EPA to develop a Risk Management Plan for preventing and responding to accidental chemical releases [2]. The maps examine whether areas with higher RMP facility proximity overlap with communities that may have greater environmental justice vulnerability based on demographic characteristics.

Data

The analysis uses an EJSCREEN dataset originally downloaded from the EPA website. The demographic indicators in EJSCREEN are based on the U.S. Census Bureau's American Community Survey (ACS) 2017–2021 5-year Summary.

Demographic and Environmental Indicators:

- Percentage of people of color (`PEOPCOLORPCT`)
- Limited English-speaking households percentile (`P_LINGISOPCT`)
- RMP facility percentile (`P_PRMP`)

Methods

EJSCREEN data were filtered to Monterey County and to Census block groups with a reported population. Coordinates for selected communities in Monterey County were stored in a simple feature layer and transformed to match the Coordinate Reference System (CRS) of the EJSCREEN dataset.

For the limited English-speaking households map, Census block groups were symbolized by `P_LINGISOPCT`, and Census block groups with `P_PRMP >= 90` were outlined to identify areas with high RMP facility proximity.

For the POC map, Census block groups were divided into four categories based on whether they met the selected thresholds.

- High RMP facility proximity 90th percentile and above (`P_PRMP >= 90`)
- High percentage of people of color 65% or above (`PEOPCOLORPCT >= 0.65`)

The 90th percentile threshold was selected as a more conservative cutoff, informed by EJSCREEN documentation that identifies the 80th percentile as a level that may warrant further review [3].

The 65% threshold was also selected as a more conservative cutoff. An analysis of the 2020 Census conducted by the Brookings Institution [4] defines a group as highly represented if its share of the area population is larger than its share of the national population. The population of POC in the United States is above 40% [4].

- High RMP facility proximity and high percentage of POC and

  (`P_PRMP >= 90 & PEOPCOLORPCT >= 0.65`)

- High percentage of POC only

  (`PEOPCOLORPCT >= 0.65 & P_PRMP < 90`)

- High proportion of RMP facility proximity

  (`P_PRMP >= 90 & PEOPCOLORPCT < 0.65`)

- Neither high POC percentage or high RMP facility proximity

  (`PEOPCOLORPCT < 0.65 & P_PRMP < 90`)

```
# Reading in data
ejscreen <- sf::st_read(here::here(
  "data",
  "ejscreen",
  "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"
))
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
   `/Users/veronicamoran/Documents/MEDS/GitHub/eds223/eds223-hw1/data/ejscreen/
EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb'
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
```

```
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
# Filtering to Monterey County
mc <- ejscreen %>%
  dplyr::filter(CNTY_NAME %in% c("Monterey County"))

# Checking CRS
st_crs(mc)
```

```
Coordinate Reference System:
  User input: WGS 84 / Pseudo-Mercator
  wkt:
PROJCRS["WGS 84 / Pseudo-Mercator",
    BASEGEOGCRS["WGS 84",
        ENSEMBLE["World Geodetic System 1984 ensemble",
            MEMBER["World Geodetic System 1984 (Transit)"],
            MEMBER["World Geodetic System 1984 (G730)"],
            MEMBER["World Geodetic System 1984 (G873)"],
            MEMBER["World Geodetic System 1984 (G1150)"],
            MEMBER["World Geodetic System 1984 (G1674)"],
            MEMBER["World Geodetic System 1984 (G1762)"],
            MEMBER["World Geodetic System 1984 (G2139)"],
            MEMBER["World Geodetic System 1984 (G2296)"],
            ELLIPSOID["WGS 84",6378137,298.257223563,
                LENGTHUNIT["metre",1]],
            ENSEMBLEACCURACY[2.0]],
        PRIMEM["Greenwich",0,
            ANGLEUNIT["degree",0.0174532925199433]],
        ID["EPSG",4326]],
    CONVERSION["Popular Visualisation Pseudo-Mercator",
        METHOD["Popular Visualisation Pseudo Mercator",
            ID["EPSG",1024]],
        PARAMETER["Latitude of natural origin",0,
            ANGLEUNIT["degree",0.0174532925199433],
            ID["EPSG",8801]],
        PARAMETER["Longitude of natural origin",0,
            ANGLEUNIT["degree",0.0174532925199433],
            ID["EPSG",8802]],
        PARAMETER["False easting",0,
            LENGTHUNIT["metre",1],
            ID["EPSG",8806]],
```

```
        PARAMETER["False northing",0,
            LENGTHUNIT["metre",1],
            ID["EPSG",8807]]],
    CS[Cartesian,2],
        AXIS["easting (X)",east,
            ORDER[1],
            LENGTHUNIT["metre",1]],
        AXIS["northing (Y)",north,
            ORDER[2],
            LENGTHUNIT["metre",1]],
    USAGE[
        SCOPE["Web mapping and visualisation."],
        AREA["World between 85.06°S and 85.06°N."],
        BBOX[-85.06,-180,85.06,180]],
    ID["EPSG",3857]]
```

```
# Filtering to columns of interest
mc <- mc |>
  filter(ACSTOTPOP > 0) |>
  select(
    ID,
    ACSTOTPOP,
    AREALAND,
    P_PRMP,
    PEOPCOLORPCT,
    P_LINGISOPCT,
  )
```

```
# Creating st_point for cities with higher populations in Monterey County

salinas_p <- st_point(c(-121.632, 36.694))
monterey_p <- st_point(c(-121.880, 36.592))
seaside_p <- st_point(c(-121.819, 36.622))
marina_p <- st_point(c(-121.789, 36.682))
soledad_p <- st_point(c(-121.322, 36.430))
greenfield_p <- st_point(c(-121.245, 36.325))
kc_p <- st_point(c(-121.127, 36.216))
cbts_p <- st_point(c(-121.922, 36.554))
cv_p <- st_point(c(-121.726, 36.484))
bradley_p <- st_point(c(-120.869, 35.863))
ml_p <- st_point(c(-121.785, 36.805))
pajaro_p <- st_point(c(-121.741, 36.902))
```

```
castroville_p <- st_point(c(-121.752, 36.766))
chualar_p <- st_point(c(-121.516, 36.570))
bs_p <- st_point(c(-121.808, 36.270))
sl_p <- st_point(c(-121.021, 36.128))
sa_p <- st_point(c(-120.906, 36.020))
jolon_p <- st_point(c(-121.176, 35.971))
spreckels_p <- st_point(c(-121.647, 36.622))
spence_p <- st_point(c(-121.554, 36.623))
gonzalez_p <- st_point(c(-121.439, 36.516))
ch_p <- st_point(c(-121.934, 36.507))
hc_p <- st_point(c(-121.869, 36.534))
rtg <- st_point(c(-121.804, 36.527))
pb_p <- st_point(c(-121.944, 36.587))
pg_p <- st_point(c(-121.920, 36.619))
cdt_p <- st_point(c(-121.765, 36.569))
sb_p <- st_point(c(-121.708, 36.569))
sv_p <- st_point(c(-121.699, 36.603))
eg_p <- st_point(c(-121.735, 36.654))
prunedale_p <- st_point(c(-121.655, 36.803))
aromas_p <- st_point(c(-121.664, 36.865))
pine_p <- st_point(c(-121.150, 36.181))
cvh_p <- st_point(c(-121.937, 36.482))


# Adding coordinate reference system

mc_points <- st_sfc(
  c(
    salinas_p,
    monterey_p,
    seaside_p,
    marina_p,
    soledad_p,
    greenfield_p,
    kc_p,
    cbts_p,
    cv_p,
    bradley_p,
    ml_p,
    pajaro_p,
    castroville_p,
    chualar_p,
    bs_p,
```

```
    sl_p,
    sa_p,
    jolon_p,
    spreckels_p,
    spence_p,
    gonzalez_p,
    ch_p,
    rtg,
    pb_p,
    pg_p,
    cdt_p,
    sb_p,
    sv_p,
    eg_p,
    prunedale_p,
    aromas_p,
    pine_p,
    cvh_p
  ),
  crs = 4326
)
```

```
# Transforming the CRS of Monterey county cities dataframe to match the CRS of
the Monterey data
mc_cities <- st_transform(mc_points, crs = st_crs(mc))
```

```
# Verifying CRS match
if (st_crs(mc_cities) == st_crs(mc)) {
  print("it's a match!")
} else {
  print("still not a match")
}
```

```
[1] "it's a match!"
```

```
# Linguistic isolation and high RMP proximity map
# Limited English-speaking households

high_h <- mc |>
  filter(P_PRMP >= 90)
```

```r
# Filtering for racial demographics and RMP facility proximity based on the selected thresholds

# High RMP facility proximity and high percentage of POC
poc_hazards <- filter(mc, P_PRMP >= 90 & PEOPCOLORPCT >= 0.65)

# High percentage of POC only
poc <- filter(mc, PEOPCOLORPCT >= 0.65 & P_PRMP < 90)

# High RMP facility proximity only
hazards <- filter(mc, P_PRMP >= 90 & PEOPCOLORPCT < 0.65)

# Neither high POC percentage nor high RMP facility proximity
low <- filter(mc, PEOPCOLORPCT < 0.65 & P_PRMP < 90)
```

Results

Census block groups with high proximity to RMP facilities above 90th percentile are located in northern Monterey County. Many of these Census block groups are above the 60th percentile for limited English-speaking households.

```r
map_in <-
  tm_shape(mc, bbox = st_bbox(poc_hazards)) +
  tm_polygons(
    fill = "P_LINGISOPCT",
    legend.show = FALSE
  ) +
  tm_shape(mc) +
  tm_borders(lwd = .3) +
  tm_shape(high_h) +
  tm_borders(
    col = "orangered3",
    lwd = 1.25
  ) +
  tm_shape(mc_cities) +
  tm_bubbles(
    size = .35,
    fill = "deepskyblue1",
    col = "black"
  ) +
  tm_scalebar(position = c("left", "bottom")) +
  tm_compass(position = c("right", "top")) +
```

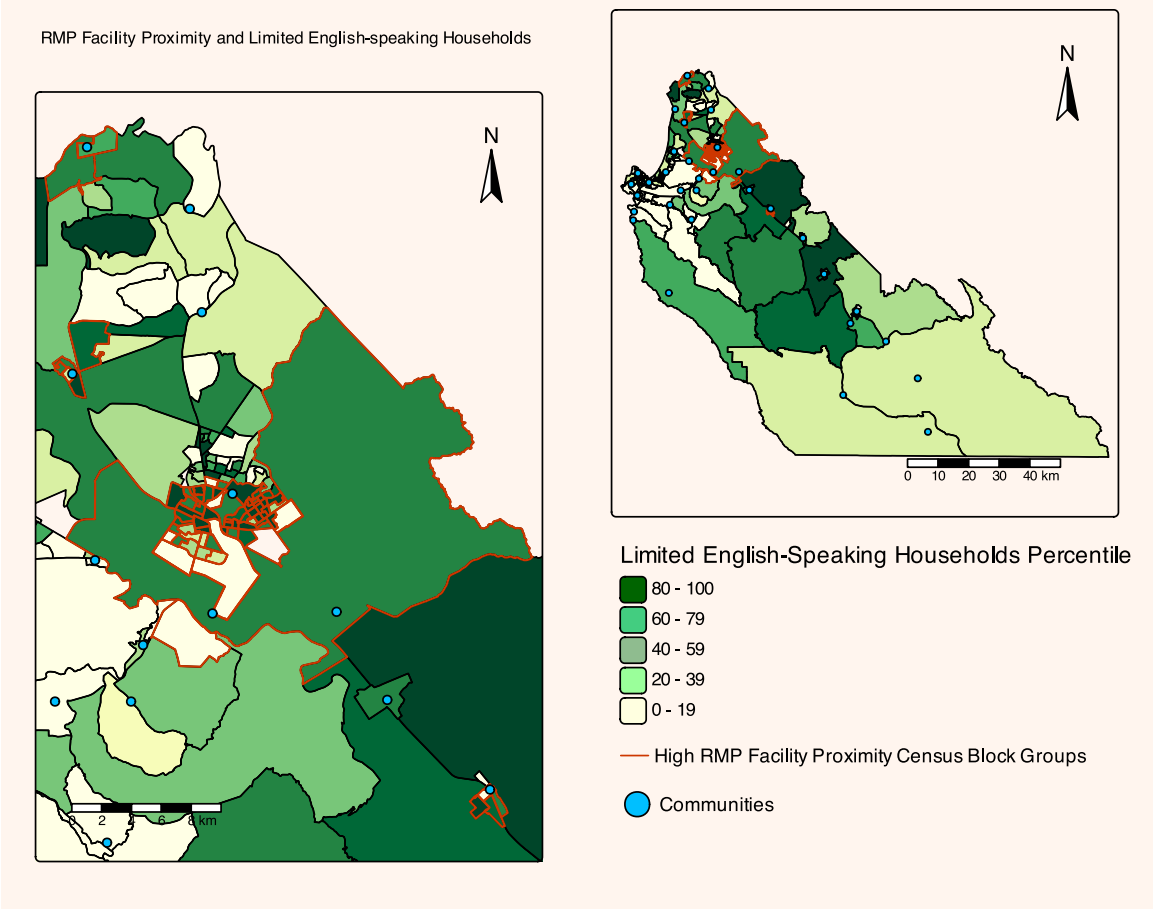
```
  tm_layout(
   main.title = "RMP Facility Proximity and Limited English-speaking Households",
    text.fontfamily = "sans",
    style = "natural",
    bg.color = "seashell",
    outer.bg = TRUE,
    outer.bg.color = "seashell",
    frame = TRUE,
    legend.text.size = 0.9,
    legend.title.size = 0.7
  )

map_out <-
  tm_shape(mc) +
  tm_polygons(
    fill = "P_LINGISOPCT",
    fill.scale = tm_scale_intervals(
      style = "fixed",
      breaks = c(0, 20, 40, 60, 80, 100),
      values = c(
        "darkgreen",
        "seagreen3",
        "darkseagreen",
        "palegreen1",
        "lightyellow"
      )
    ),
    legend.show = FALSE
  ) +
  tm_shape(mc) +
  tm_borders(lwd = .3) +
  tm_shape(high_h) +
  tm_borders(
    col = "orangered3",
    lwd = 1.25
  ) +
  tm_shape(mc_cities) +
  tm_bubbles(
    size = .25,
    col = "black",
    fill = "deepskyblue1"
  ) +
  tm_scalebar(position = c("right", "bottom")) +
```

```
  tm_compass(position = c("right", "top")) +
  tm_add_legend(
    type = "polygons",
    title = "Limited English-Speaking Households Percentile",
    fill = c(
      "darkgreen",
      "seagreen3",
      "darkseagreen",
      "palegreen1",
      "lightyellow"
    ),
    labels = c(
      "80 - 100",
      "60 - 79",
      "40 - 59",
      "20 - 39",
      "0 - 19"
    ),
    bg.color = "seashell",
    bg.alpha = 1
  ) +
  tm_add_legend(
    type = "lines",
    title = "",
    col = "orangered3",
    labels = "High RMP Facility Proximity Census Block Groups",
    bg.color = "seashell",
    bg.alpha = 1
  ) +
  tm_add_legend(
    type = "symbols",
    title = "",
    fill = "deepskyblue1",
    col = "black",
    labels = "Communities",
    bg.color = "seashell",
    bg.alpha = 1
  ) +
  tm_layout(
    text.fontfamily = "sans",
    style = "natural",
    bg.color = "seashell",
    outer.bg = TRUE,
```

```
    outer.bg.color = "seashell",
    frame = TRUE
  )
```

```
tmap_arrange(
  map_in,
  map_out,
  ncol = 2,
  asp = 0,
  outer.margins = c(0.05, 0.06, 0.05, 0.06)
)
```



Census block groups with high RMP facility proximity overlap with populations that are at least 65% people of color. Most of these Census block groups are located in northern Monterey County. In coastal and southern Monterey County RMP facility proximity and the percentage of people of color fall below the threshold.

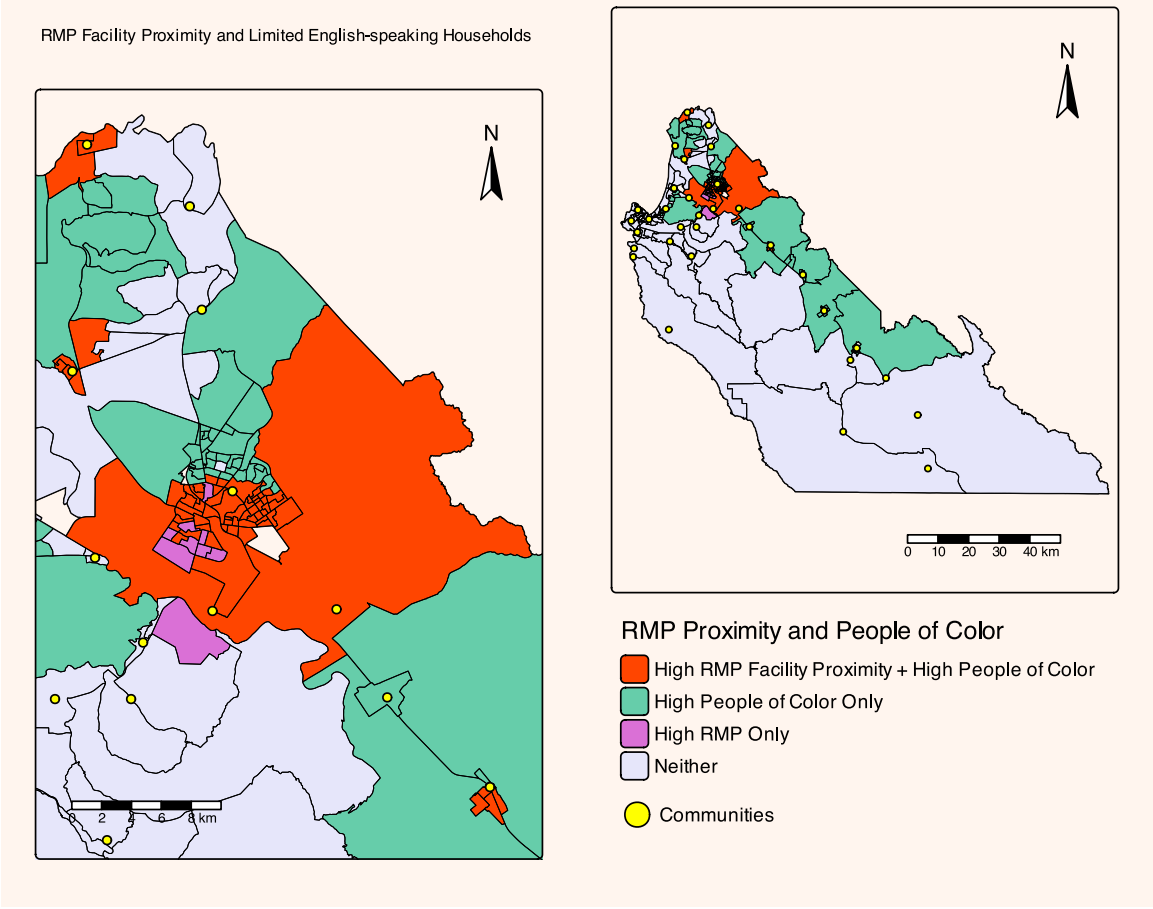
```r
# Racial demographics and high RMP facility proximity map
map_zoom <-
  tm_shape(mc, bbox = st_bbox(poc_hazards)) +
  tm_polygons(fill = NA, col = "black") +
  tm_shape(poc_hazards) +
  tm_polygons(fill = "orangered1", col = NULL) +
  tm_shape(poc) +
  tm_polygons(fill = "mediumaquamarine", col = NULL) +
  tm_shape(hazards) +
  tm_polygons(fill = "orchid", col = NULL) +
  tm_shape(low) +
  tm_polygons(fill = "lavender", col = NULL) +
  tm_shape(mc) +
  tm_borders(
    col = "black",
    lwd = .35
  ) +
  tm_shape(mc_cities) +
  tm_bubbles(size = .35, col = "black", fill = "yellow") +
  tm_scalebar(position = c("left", "bottom")) +
  tm_compass(position = c("right", "top")) +
  tm_layout(
   main.title = "RMP Facility Proximity and Limited English-speaking Households",
    text.fontfamily = "sans",
    style = "natural",
    bg.color = "seashell",
    frame = TRUE,
    outer.bg.color = "seashell",
    legend.text.size = 0.9,
    legend.title.size = 0.7
  )

# Racial demographics and high RMP facility proximity map

map_far <-
  tm_shape(mc) +
  tm_polygons(fill = NA, col = "black") +
  tm_shape(poc_hazards) +
  tm_polygons(fill = "orangered1", col = NULL) +
  tm_shape(poc) +
  tm_polygons(fill = "mediumaquamarine", col = NULL) +
  tm_shape(hazards) +
  tm_polygons(fill = "orchid", col = NULL) +
```

```
  tm_shape(low) +
  tm_polygons(fill = "lavender", col = NULL) +
  tm_shape(mc) +
  tm_borders(
    col = "black",
    lwd = .35
  ) +
  tm_shape(mc_cities) +
  tm_bubbles(size = .25, col = "black", fill = "yellow") +
  tm_scalebar(position = c("right", "bottom")) +
  tm_compass(position = c("right", "top")) +
  tm_add_legend(
    type = "polygons",
    title = "RMP Proximity and People of Color",
    fill = c("orangered1", "mediumaquamarine", "orchid", "lavender"),
    labels = c(
      "High RMP Facility Proximity + High People of Color",
      "High People of Color Only",
      "High RMP Only",
      "Neither"
    ),
    bg.color = "seashell",
    bg.alpha = 1
  ) +
  tm_add_legend(
    type = "symbols",
    title = "",
    fill = "yellow",
    col = "black",
    labels = "Communities",
    bg.color = "seashell",
    bg.alpha = 1
  ) +
  tm_layout(
    text.fontfamily = "sans",
    style = "natural",
    bg.color = "seashell",
    frame = TRUE,
    outer.bg.color = "seashell"
  )
```

```
tmap_arrange(
  map_zoom,
  map_far,
  ncol = 2,
  asp = 0,
  outer.margins = c(0.05, 0.06, 0.05, 0.06)
)
```



Conclusion

These maps highlight potential environmental justice concerns related to Risk Management Program (RMP) facility proximity. The Census block groups with RMP facility proximity above the 90th percentile are located in northern Monterey County. Many Census block groups with high RMP facility proximity have limited English-speaking household percentiles of 60 or higher and populations that are at least 65% people of color. In comparison, coastal and southern Census block groups generally have lower RMP facility proximity, lower percentiles of limited English-speaking households and lower percentages of people of color.

These patterns do not measure exposure or health effects, but they identify areas where higher RMP facility proximity overlaps with potentially vulnerable communities. Populations in northern Monterey County, particularly in and around Salinas, may experience greater environmental injustices related to RMP facility proximity. Further study could examine environmental justice concerns in Monterey County by including factors not captured in the EJSCREEN dataset, such as the population of immigrant farmworkers living in the area. A limitation of this type of analysis is the potential lack of representative or accurate data. Immigrants may be hesitant to share personal information due to fear, privacy concerns, or lack of trust in people or institutions outside of their community.

References

[1]. U.S. Environmental Protection Agency. (2021). *Environmental Justice Screening and Mapping Tool: What is EJSCREEN.* (https://19january2021snapshot.epa.gov/ejscreen/what-ejscreen_.html) [Accessed: October 7, 2026]

[2]. U.S. Environmental Protection Agency. (2025, December 4). *Risk Management Program (RMP) Rule Overview.* (https://www.epa.gov/rmp/risk-management-program-rmp-rule-overview) [Accessed: October 7, 2026]

[3]. U.S. Environmental Protection Agency (EPA), 2023. *EJScreen Technical Documentation.*

[4]. Frey, W. H. (2021, August 13). *New 2020 census results show increased diversity countering decade-long declines in America's white and youth populations.* Brookings Institution. (https://www.brookings.edu/articles/new-2020-census-results-show-increased-diversity-countering-decade-long-declines-in-americas-white-and-youth-populations/) [Accessed: October 9, 2026]